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PERFORMANCE SPECIFICATION
VIDEO PREAMPLIFIER, AM-6923/UA

This performance specification is approved for use by USACECOM, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This performance specification covers one type of common module, Video Preamplifier, AM-6923/UA, herein referred to as the "Preamplifier".

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2). Handbooks are for guidance only and therefore are not mandatory.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-LC-LEO-E-EP, Fort Monmouth, NJ. 07703 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5855

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HANDBOOKS

DEPARTMENT OF DEFENSE

MIL-HDBK-781	- Reliability Test Methods, Plans, and Environments for Engineering Development, Qualification, and Production.
MIL-HDBK-454	- Standard General Requirements for Electronic Equipment.

(Unless otherwise indicated, copies of the above specifications, standards and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094).

2.2.2 Other Government documents, drawings and publications. The following other government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2.i).

DRAWINGS

USA CECOM

SM-D-773663	- Preamplifier, Video AM-6923/UA
SM-D-773665	- Schematic Diagram, IR Video Preamp Assy
SM-C-771821	- Preamplifier PWB, Outline Dimensions

(Copies of documents, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2.i).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/IPC-S-815A	- General Requirements for Soldering Electronic Interconnections

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(Application for copies should be addressed to the Institute for Interconnecting and Packaging Electronic circuits, 3451 Church Street, Evanston, Illinois, 60203).

AMERICAN SOCIETY FOR QUALITY CONTROL

ANSI/ASQC Z 1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Application for copies should be addressed to the American Society for Quality Control, P.O. Box 3005, 611 East Wisconsin Avenue, Milwaukee, Wisconsin 53202).

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract (see 6.2.b) the contractor shall furnish first article units for inspection (4.2).

3.2 Materials. The contractor shall select materials which shall be fully capable of meeting all of the operational and environmental requirements specified herein. The materials specified in the applicable drawings are recommended, but are not mandatory. Selection criteria of the class, grade or type part shall be that the material will be able to perform its intended function when it is assembled. Verification of the supplier meeting the overall performance requirements shall be the governing acceptance standard. Recovered materials shall be used to the maximum extent possible.

3.3 Description. The Preamplifier is a specialized circuit for amplifying the signal from a detector array in an infrared system.

3.4 Design. The Preamplifier shall meet the form, fit, function and interface requirements specified herein.

3.4.1 Weight. The weight of the Preamplifier shall be 0.16, (+0.02 -0.03) pound as measured with a scale having an accuracy of 99% or better.

3.4.2 Components. The Preamplifier shall consist of twenty channels of amplification provided on each printed wiring board assembly.

3.4.3 Dimensions. Dimensions and dimensional tolerances shall be in accordance with drawing SM-D-773663 for maximum overall height of components from the printed wiring board. Dimensions of the printed wiring board shall be in accordance with drawing

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SM-C-771821 to ensure proper physical interface with existing hardware. Pinouts to the connectors shall be in accordance with drawing SM-D-773665 to ensure proper electrical interconnections.

3.5 Performance characteristics.

3.5.1 Equivalent input/burst noise voltage. Equivalent input noise, including burst noise, shall not be greater than 1.5×10^{-9} volts rms/$\sqrt{}$ Hz.

3.5.2 Equivalent input 1/f noise. Equivalent input 1/f noise shall not be greater than 10×10^{-9} volts / $\sqrt{}$ Hz measured at 100 Hz.

3.5.3 Channel crosstalk. Channel crosstalk (19 channels into 1 channel) shall be less than -30 dB with an input voltage e_{in} of 1 millivolts (mV) rms at 1 kHz.

3.5.4 Channel-to-channel tracking. The voltage gain of the Preamplifier channels shall track within ±5 percent of the average voltage gain over the temperature range of -54°C to +71°C.

3.5.5 Voltage gain. Voltage gain for each of the twenty channels shall be 71.5, ±8.5 volts per volt (V/V) at 1 kHz.

3.5.6 Signal bandwidth. The upper 3 dB frequency shall be between 76 kHz and 164 kHz (Midband gain reference: 0 dB at 1 kHz).

3.5.7 Input impedance. Input impedance shall not be less than 4,000 ohms and not greater than 6,500 ohms.

3.5.8 Output impedance. Output impedance shall not be less than 400 ohms and not greater than 700 ohms.

3.5.9 Input power. The input power demand by the Preamplifier shall not be greater than 65 milliamperes (mA) with a supply voltage of 9.75, ±0.25 volts direct current (Vdc) in the high power mode. The output of the on-board regulator shall be 3.6, ±0.4 Vdc with an input supply of 9.75, ±0.25 Vdc.

3.5.10 Recovery time. Recovery time from a 50 millisecond, 10 mV pulse input, followed by a signal of 10 milliseconds width and 0.7 mV amplitude, shall not be greater than 200 milliseconds.

3.5.11 Maximum input signal. The maximum sinewave input signal for an output signal with less than 5 percent distortion shall be 10 mV peak-to-peak (p-p) at 1 kHz.

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3.5.12 Total signal bandwidth. The upper 3 dB frequency shall be between 76 and 164 kHz. The lower 3 dB frequency shall be 4, ±2 Hz. The Preamplifier gain shall be flat (±0.5 dB of midband gain) from 30 Hz to 25 kHz (Midband gain reference: 0 dB at 1 kHz).

3.5.13 Voltage gain drift. The average gain of the 20 channels on each board shall not differ by more than ±10 percent from the average gain at ambient temperature of +23°C, ±5°C, over the operating temperature range of -54°C to +71°C.

3.6 Environmental conditions.

3.6.1 Temperature shock. The Preamplifier shall not be damaged (see 6.3.1) by sudden changes in temperature between -54°C and +95°C.

3.6.2 High temperature. The Preamplifier shall not be damaged by storage to +95°C or continuous operation to +71°C, and shall operate for a minimum of 30 minutes at +95°C.

3.6.3 Low temperature. The Preamplifier shall not be damaged by continuous operation to -54°C or storage to -62°C.

3.6.4 Shock. The Preamplifier shall not be damaged by high intensity shocks of 100 g's peak amplitude with 11 milliseconds duration and by bench handling tests.

3.6.5 Vibration. The Preamplifier shall not be damaged by vibration over the frequency spectrum at the specified "g" levels and amplitudes shown in Figure 1.

3.7 Reliability. The Preamplifier shall have a mean time between failure (MTBF) of at least 26,000 hours, θ_0.

3.8 Coating. Unless otherwise specified, the Preamplifier and its components and parts shall be cleaned and coated in accordance with the applicable drawings.

3.9 Preconditioning. Each Preamplifier shall be subjected to at least a 20-cycle preconditioning environment as specified in Figure 2 prior to Group A inspection. For exception to this requirement see 6.2.f.

3.10 Soldering. Unless otherwise specified, soldering shall be in accordance with ANSI/IPC-S-815A.

3.11 Nameplates and product marking. The Preamplifier, parts, components, subassemblies and assemblies thereof shall be marked for identification and reference designation markings. For guidance, use MIL-HDBK-454. Markings shall include serial number, cage code of buying activity and contract number.

3.12 Workmanship. All electronic parts, components, assemblies and sub-assemblies shall be free of smudges, excess solder, metal chips or the existence of any foreign material on any

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surface. Bearing assemblies shall be free from rust, dirt or tool marks. Wires and integrated circuitry shall be protected from contact with rough or irregular surfaces and be shielded from shorting.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.4).

4.2 First article inspection. When specified in the contract, first article inspection shall be performed by the contractor on a minimum of 3 sample units. First article reliability (4.2.3) is not required unless specified in the contract.

4.2.1 Inspections. All materials, parts, processes and assemblies shall be visually examined for conformance to the applicable specification or drawing and Table I inspections. Inspection shall be made using sampling per ANSI/ASQC Z 1.4.

TABLE I. Visual inspections.

Inspection requirement	Requirement paragraph	Verification paragraph
Any part or component missing or damaged	3.4.2	4.2.1
Weight not as specified	3.4.1	4.2.1
Soldering not as specified	3.10	4.2.1
Marking not as specified	3.11	4.2.1
Workmanship not as specified	3.12	4.2.1
Coating not as specified	3.8	4.2.1
Dimensions not as specified	3.4.3	4.2.1
Length	3.4	4.2.1
Width	3.4	4.2.1
Height, max	3.4	4.2.1
Solder/lead protrusion	3.4	4.2.1
Pins, connector location	3.4	4.2.1
Pins, guide location	3.4	4.2.1

*In ANSI/ASQC Z 1.4 use inspection level S-3 or objective quality evidence per ASQC Z 1.4.

4.2.2 Tests. Upon successful completion of the inspections specified in 4.2.1, perform the tests of Table II in any order. Failure of any test shall constitute failure of first article inspection.

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TABLE II. First article inspection.

Inspection	Requirement paragraph	Test paragraph
Equivalent input/burst noise voltage	3.5.1	4.5.1
Equivalent input 1/f noise	3.5.2	4.5.2
Channel crosstalk	3.5.3	4.5.3
Channel to channel tracking	3.5.4	4.5.4
Voltage gain	3.5.5	4.5.5
Signal bandwidth	3.5.6	4.5.6
Input impedance	3.5.7	4.5.7
Output impedance	3.5.8	4.5.8
Input power	3.5.9	4.5.9
Recovery time	3.5.10	4.5.10
Maximum input signal	3.5.11	4.5.11
Total signal bandwidth	3.5.12	4.5.12
Voltage gain drift	3.5.13	4.5.13
Temperature shock	3.6.1	4.6.1
High temperature	3.6.2	4.6.2
Low temperature	3.6.3	4.6.3
Shock	3.6.4	4.6.4
Vibration	3.6.5	4.6.5

4.2.3 First article reliability. The lower test MTBF (θ_1) shall be demonstrated in accordance with the following subparagraphs.

4.2.3.1 Inspection level.

- Temperature: +50°C, (+5 -0)°C
- Temperature cycling: None
- Vibration: 2.2 g's, ±10 percent peak acceleration value at any nonresonant frequency between 20 and 60 Hz measured at the mounting points of the equipment. The duration of vibration shall be at least 10 minutes during each hour of equipment operating time. The vibration shall be normal to the plane of the printed circuit board.
- Equipment on-off cycling: Turn the equipment ON and let the temperature stabilize, hold for 3 hours. Then turn the equipment OFF and let the temperature stabilize. This cycle shall continue throughout the test.

4.2.3.2 Parameters. The discrimination ratio θ_0/θ_1 shall be 3.

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4.2.3.3 Failure criteria and test length. The tests of Table III shall be performed at least once prior to and once after the tests of 4.2.3.1. Additional testing of Table III may be selected by the contractor. The results of all Table III tests and any physical damage shall be used to determine the success or failure of the reliability test. In the event of a Table III test failure, the time interval between the preceding successful test and the failure shall not contribute to the total cumulative hours. Testing shall continue until the total unit hours, together with the total count of relevant equipment failures, permit either an accept or reject decision in accordance with Figure 6. Only equipment "ON" time may contribute to the total unit hours. No single equipment's "ON" time shall be less than one-half the average operating time of all equipment on test.

4.2.4 Disposition of first article samples. First article samples shall not be considered as part of the procurement quantities (see 6.2.g).

4.3 Inspection procedures for quality assurance provisions. Unless otherwise specified herein, the Preamplifier shall be operated in high power mode at an ambient temperature of +23°C, ±5°C. The input power shall be in accordance with 3.5.9. Unless otherwise specified, all test setup and load resistor tolerances shall be ±5 percent.

4.4 Conformance inspection.

4.4.1 Group A inspection. Group A inspection shall be conducted at an ambient temperature of +23°C, ±5°C on all Preamplifiers following the preconditioning of 3.9 and the inspection of 4.2.1. Group A tests listed in Table III may be performed in any order. Failure of any test shall be cause for rejection of that unit.

TABLE III. Group A inspection.

Inspection	Requirement paragraph	Test paragraph
Equivalent input/burst noise voltage	3.5.1	4.5.1
Voltage gain	3.5.5	4.5.5
Signal bandwidth	3.5.6	4.5.6
Input power	3.5.9	4.5.9

4.4.2 Group B inspection. Not required.

4.4.3 Group C inspection. Group C inspection shall be conducted on Preamplifiers selected from units which have passed the tests in 4.4.1. The samples shall be tested in accordance with the inspections listed in Table IV. Group C tests listed in table IV may be performed in any order. Group C inspection samples shall be selected randomly. The sampling plan is as follows:

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MONTHLY LOT SIZE	NUMBER OF SAMPLE UNITS MONTHLY
1-10	1
11-49	2
50-99	4
100-299	5
300-499	6
500-999	8
1000 or more	10

TABLE IV. Group C inspection.

Inspection	Requirement paragraph	Test requirement
Equivalent input 1/f noise	3.5.2	4.5.2
Channel-to-channel tracking	3.5.4	4.5.4
Vibration	3.6.5	4.6.5
Input impedance	3.5.7	4.5.7
Output impedance	3.5.8	4.5.8
Recovery time	3.5.10	4.5.10
Maximum input signal	3.5.11	4.5.11
Voltage gain drift	3.5.13	4.5.13

4.4.3.1 Group C failures. Actions required relative to Group C failures shall be specified in the contract (see 6.2.d).

4.4.3.2 Disposition of Group C samples. Group C samples shall be accepted on contract subsequent to the tests of Table III.

4.5 Methods of inspection.

4.5.1 Equivalent input/burst noise voltage. Equivalent input/burst noise voltage shall be determined by measuring the wideband rms noise voltage output with a true rms voltmeter when the input to the Preamplifier is terminated with a 51-ohm resistor. Use the following equation:

Where: $e_{noise}/\sqrt{Hz} = \{[\, e_{out} \div (A_V \times (NBW)^{1/2})]^2 - (V_{RS})^2\}^{1/2}$

$e_{noise}/\sqrt{Hz}$ = equivalent input noise

e_{out} = maximum rms noise voltage

A_V = voltage gain

NBW = noise bandwidth = π /2 x signal bandwidth

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R_S = 51 ohms (equivalent detector resistance)

$V_{RS} = (4kTR_S)^{1/2}$

k = Boltzman constant (1.37×10^{-23} joule/K)

T = absolute temperature, K

The test methods shall consist of one of two alternative methods as follows:

a. Using a digital true rms voltmeter, a minimum of 5 consecutive one-second integration times or 10 consecutive 500-msec integration times shall be taken and the maximum rms noise output voltage recorded.

b. Using an analog true rms voltmeter and a dc-coupled oscilloscope monitoring e_{out}, the average total equivalent input noise observed for a minimum of 5 seconds shall be not greater than 1.5 nanovolts/ $\sqrt{Hz}$ as computed in the preceding equation. The total output noise (p-p) on the oscilloscope shall be not greater than $6A_v$ microvolts, where A_v is the overall voltage gain of the preamplifier under test.

Since the noise output of the Preamplifier will be very low, it may fall under the minimum threshold level of the rms voltmeter. The measurement may then be performed using a cascaded second low-noise amplifier. This amplifier must have a bandwidth that is at least two times as great as the Preamplifier bandwidth that is being measured. Source resistance noise is removed from the total rms output noise by quadrature subtraction. Failure to meet the requirements of 3.5.1 shall constitute failure of this test.

4.5.2 Equivalent input 1/f noise. Equivalent input 1/f noise voltage shall be measured with a wave analyzer at 100 Hz, with Δf = 10 Hz, and the input to the preamplifier is terminated with a 51 ohm resistor. Using voltage gain as measured per 3.5.5, equivalent input 1/f noise can be calculated with the following equation:

$$e_{noise}/\sqrt{Hz} = \frac{e_{noise\ out}}{[A_v(\Delta f)^{1/2}]}$$

Where $e_{noise\ out}$ = wave analyzer output at 100Hz

A_v = voltage gain = V_{out}/V_{in}

Δf = 10 Hz (effective noise bandwidth)

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Since the noise output of the preamplifier may be less than the minimum threshold of the wave analyzer, the measurement may be performed using a cascaded second low-noise wideband amplifier. Failure to meet the requirements of 3.5.2 shall constitute failure of this test.

4.5.3 Channel crosstalk. Channel crosstalk (19 channels into 1 channel) shall be measured with an input voltage V_{in} of 1 mV rms at 1 kHz to any 19 channels and with the input to the channel of interest terminated into 51 ohms. Using the following equation, calculate the channel crosstalk:

$$\text{Channel crosstalk} = 20 \log (19\ V_{out\ 20}) \div (V_{out,1}+V_{out\ 2}+\text{----}V_{out\ 19})$$

All 20 channels shall be tested. Failure to meet the requirements of 3.5.3 shall constitute failure of this test.

4.5.4 Channel-to-channel tracking. Voltage gain shall be measured per 4.5.5. Temperature tests shall be performed using a temperature chamber and loads of 51 ohms input and 10,000 ohms output. Test leads shall be run outside the chamber for monitoring each channel. Voltage gain of each channel in defined by the following equation:

$$(0.95)\ A_T\ \frac{A_{I,T=25°C}}{A_{T=25°C}} \leq A_{I,T} \leq (1.05)\ A_T\ \frac{A_{I,T=25°C}}{A_{T=25°C}}$$

Where $A_{I,T}$ = voltage gain of channel I at temperature T

A_T= average gain of 20 channels at temperature T

A_T = 25°C average gain of 20 channels at temperature T = 25°C

$$A_T = \sum_{I=1}^{I=20} A_{I,T} \div 20$$

Note: Brackets = room temperature normalization factor.

4.5.5 Voltage gain. Voltage gain shall be measured using a signal generator and oscilloscope or rms voltmeter. The input to the Preamplifier shall be terminated with a 51-ohm resistor. With a measured peak voltage input e_{in} of 1.4 mV (1 mV rms), and the signal generator set at the Preamplifier midband frequency of 1 kHz, the amplifier rms voltage output e_{out} shall be measured. The midband gain of the amplifier shall be:

$$A_v = \frac{e_{out}}{e_{in}}$$

where e_{in} = rms input voltage
e_{out} = rms output voltage

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Failure to meet the requirements of 3.5.5 shall constitute failure of this test.

4.5.6 Signal bandwidth. Signal bandwidth shall be measured using the test setup of 4.5.5. The upper 3 dB frequency of the amplifier shall be measured by increasing the frequency of the signal generator beyond the midband reference of 1 kHz until the output voltage measured is down 3 dB. Failure to meet the requirements of 3.5.6 shall constitute failure of this test.

4.5.7 Input impedance. Input impedance shall be measured using a signal generator with a 50 ohm source resistance and a 50 microvolt rms (70 microvolt peak) sinusoidal signal set at the Preamplifier midband frequency of 1 kHz. The input and output voltages shall be monitored simultaneously. The output signal shall be attenuated by inserting a resistor in series with the input signal. The impedance shall be the value of the series resistance required to attenuate the output to 50 percent of the initial value. Failure to meet the requirements of 3.5.7 shall constitute failure of this test.

4.5.8 Output impedance. The output impedance shall be measured using the test setup of 4.5.7. An input signal of 50 microvolts rms (70 microvolts peak at 1 kHz) shall be injected directly into the Preamplifier while the no-load resistance output is monitored. The output signal shall be attenuated by adding a load resistor to the output. The output impedance shall be the value of the load resistor required to attenuate the signal to 50 percent of its unloaded value. Failure to meet the requirements of 3.5.8 shall constitute failure of this test.

4.5.9 Input power. Input power shall be determined by inserting a current meter in series with the power supply and Preamplifier under test. Supply voltage shall be set per 3.5.9 for high power mode. Failure to meet the requirements of 3.5.9 shall constitute failure of this test.

4.5.10 Recovery time. Recovery time shall be measured using a pulse generator and source resistance of 51 ohms and a Preamplifier load of 10,000 ohms. The pulse V_1 (T_1) shall be injected into the Preamplifier input, with an amplitude V_1 of 10 mV and a width T_1 of 50 milliseconds followed by pulse V_3 (T_3), with an amplitude of 0.7 mV and a duration T_3 of 10 milliseconds. The Preamplifier output from V_1 (T_1) is V'_1, and the output from V_3 (T_3) is V'_3. The Preamplifier input shall be monitored continuously on channel A of a dual channel oscilloscope. The Preamplifier output shall be monitored on channel B and synchronized with channel A. The recovery time shall be determined as follows: Interval T_2 measured on channel A, shall be reduced until the trailing edge of output pulse V_1 measured on channel B and the leading edge of output pulse V_3' meet. Interval T_2 is the recovery time (see Figure 3). Failure to meet the requirements of 3.5.10 shall constitute failure of this test.

4.5.11 Maximum input signal. Maximum sinewave input for an undistorted output shall be measured at the output of the Preamplifier. The output distortion shall not be greater than 5 percent with an input of 10 mV p-p at 1 kHz. The Preamplifier load shall be 10,000, ±2,500 ohms, which is equivalent to the postamplifier/control driver input impedance. This measurement shall be made using a signal generator and a distortion analyzer. The input to the Preamplifier

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shall be shunted with a 51-ohm resistor. Failure to meet the requirements of 3.5.11 shall constitute failure of this test.

4.5.12 Total signal bandwidth. Signal bandwidth shall be measured using the test setup of 4.5.5. The upper 3 dB frequency of the Preamplifier shall be measured by increasing the frequency of the signal generator beyond the midband reference of 1 kHz until the output voltage measured is down 3 dB. The lower 3 dB frequency of the Preamplifier shall be measured by decreasing the frequency of the signal generator below the midband reference of 1 kHz until the output voltage is down 3 dB. Preamplifier flatness shall be determined by measuring the frequencies above and below the midband reference of 1 kHz. Failure to meet the requirements of 3.5.12 shall constitute failure of this test.

4.5.13 Voltage gain drift. Voltage gain shall be measured using the methods described in 4.5.5. Temperature tests shall be performed using a temperature chamber and loads of 51 ohms input and 10,000 ohms output. Test leads shall be run outside the chamber for monitoring each channel. This test may be performed in conjunction with temperature tests 4.6.1, 4.6.2 and 4.6.3. Failure to meet the requirements of 3.5.13 shall constitute failure of this test.

4.6 Environmental tests. Environmental testing may be performed in accordance with the methods and procedures recommended in 6.3. The contractor may develop and use an alternative, (equivalent) test plan following Government review and approval. When the contractor's test plan deviates from the recommended test procedures, the contractor shall be required to show where his own tests and methods for verifying the specified performance are equivalent. Also, measurements taken by the contractor during verification testing of all environmental requirements shall be accomplished using known testing apparatus such as an environmental test chamber, vibration test table, pendulum shock device or similar equipment for which the technical capabilities to properly simulate extreme climatic or environmental conditions are scientifically documented.

4.6.1 Temperature shock. Verify compliance with the requirement of 3.6.1. Use of the procedures identified in 6.3 is recommended. Failure to meet the requirement shall constitute failure of the test.

4.6.2 High temperature. Verify compliance with the requirement of 3.6.2. Use of the procedures identified in 6.3, or equivalent method, is recommended. Failure to meet the requirement shall constitute failure of the test.

4.6.3 Low temperature. Verify compliance with the requirement of 3.6.3. Use of the procedures identified in 6.3, or equivalent method, is recommended. Failure to meet the requirement shall constitute failure of the test.

4.6.4 Shock. Verify compliance with the requirement of 3.6.4. Use of the procedures identified in 6.3, or equivalent method, is recommended. Failure to meet the requirement shall constitute failure of the test.

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4.6.5 Vibration. Verify compliance with the requirement of 3.6.5. Use of the procedures identified in 6.3, or equivalent method, is recommended. Failure to meet the requirement shall constitute failure of the test.

4.7 Reliability test.

4.7.1 First article reliability. The lower test MTBF (θ_1) shall be demonstrated in accordance with the following subparagraphs.

4.7.1.1 Test level.

Temperature: +50°C, (+5 -0)°C

Temperature: Cycling: None

Vibration: 2.2 g's, ±10 percent peak acceleration value at any nonresonant frequency between 20 and 60 Hz measured at the mounting points of the equipment. The duration of vibration shall be at least 10 minutes during each hour of equipment operating time. The vibration shall be perpendicular to the plane of the printed circuit board.

Equipment ON-OFF cycling: Turn the equipment ON and let the temperature stabilize, hold for 3 hours, then turn the equipment OFF and let the temperature stabilize. This cycle shall continue throughout the test.

4.7.1.2 Parameters. The discrimination ratio θ_0/θ_1 shall be 3.

4.8 Failure criteria and test length. The test of Table III shall be performed at least once prior to and once after the tests of 4.7.1.1. Additional testing of Table III may be selected by the contractor. The results of all Table III tests and any physical damage shall be used to determine the success of failure of the reliability test. In the event of a Table III test failure, the time interval between the preceding successful test and the failure shall not contribute to the total cumulative hours. Testing shall continue until the total unit hours, together with the total count of relevant equipment failures, permit either an accept or reject decision in accordance with Figure 6. Only equipment "on" time shall contribute to the total unit hours. No single equipment's "on" time shall be less than one-half the average operating time of all equipment on test.

4.9 Preconditioning. Each Preamplifier, in a non-operating mode, shall be subjected to at least 20 cycles as specified in Figure 2. Preconditioning may be performed before or after conformal coating.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2.e). When actual packaging of material is to be performed by

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DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. The Preamplifier is intended for use in infrared systems.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. First article tests are required:

(1) Time frame for submission of first article test reports are required.

(2) Time frame for approval of first article test reports are required.

(3) First article reliability testing, if required.

c. Sampling plans for Group C.

d. Necessary action by the contractor in the event of a Group C failure (see 4.4.3.1).

e. Packaging requirements.

f. The preconditioning requirement 3.9 is required for procurement of all modules to this specification made directly for the Government. Implementation of 3.9 may be waived by systems contractors ordering modules to this specification for use in the manufacture of complete systems if these systems or subassemblies thereof are subjected to equivalent preconditioning.

g. Deliver all first article samples as specified by the procuring activity's Contracting Officer.

h. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specification or preparing activity for recommended disposability methods.

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i. Issue of DoDISS to be cited in the solicitation, and if required, the specified issue of individual document referenced (see 2.2 and 2.3).

j. Environmental pollution prevention measures are contained in the packaging material specifications or provided by preparing activity for recommended disposability methods.

6.3 Definitions.

6.3.1 Damage. Damage is defined as breakage, loosening, shifting, evidence of corrosion, or failure of any finish, hardware, connection or component; and degradation of Preamplifier performance to values less than specified herein.

6.4 Environmental inspection procedures. The test methods and procedures of MIL-STD-810 listed below are recommended, but are not mandatory for determining compliance with the environmental requirements of this specification.

TABLE VI. Environmental inspections.

Inspection	Requirement paragraph	MIL-STD-810 method	MIL-STD-810 procedure	Remarks
Temperature shock	3.6.1	503	I	Non-operating mode. Step 2 use -54°C and steps 1 & 4 use +95°C.
High Temperature (see Figure 4)	3.6.2	501	I	Test using set-up of 3.5.5 during the final half hour of step 3. For step 2 use +95°C and steps 4 & 5 use +71°C. The operating tests of 3.5.4 & 3.5.13 shall be performed in lieu of Table III, during this test.
Low temperature	3.6.3	502	I	The Operating tests of 3.5.4

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(see Figure 5)				and 3.5.13 shall be performed during this test, in lieu of Table III. Step 2 use -62°C and steps 4 use -54°C.
Shock	3.6.4	516.1	IV, V	Non-operating mode.
Vibration (see Figure 1)	3.6.5	514.1	I, VII	Non-operating mode. Use Figure 1 of this specification in lieu of MIL-STD-810 . Sinusoidal cycling time shall be 120 minutes. Overall time shall be 1/6 of the cycling time at each resonance.

6.5 Subject term (key word) listing.

Detector array
infrared system

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CUSTODIAN:
Army-CR

PREPARING ACTIVITY:
Army-CR

(Project 5855-0108)

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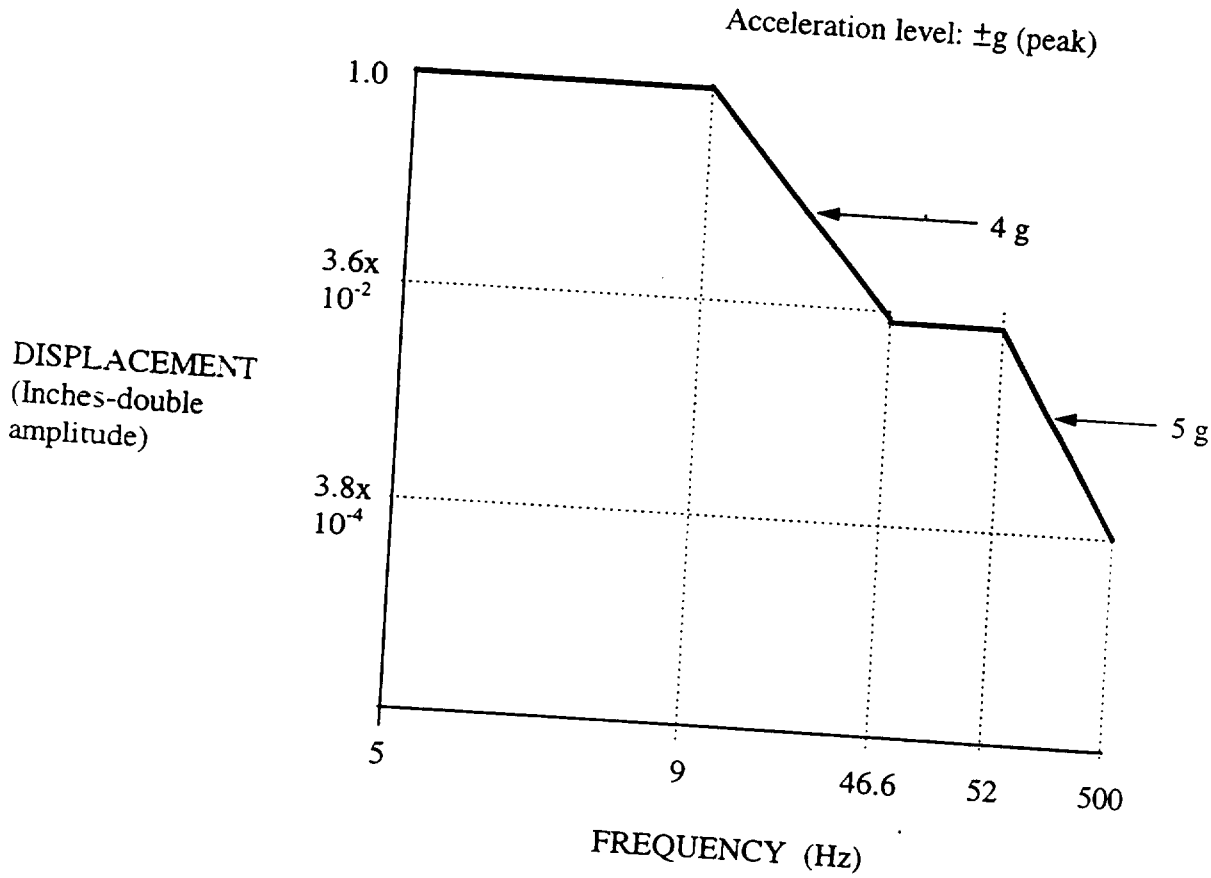


FIGURE 1. Vibration test profile.

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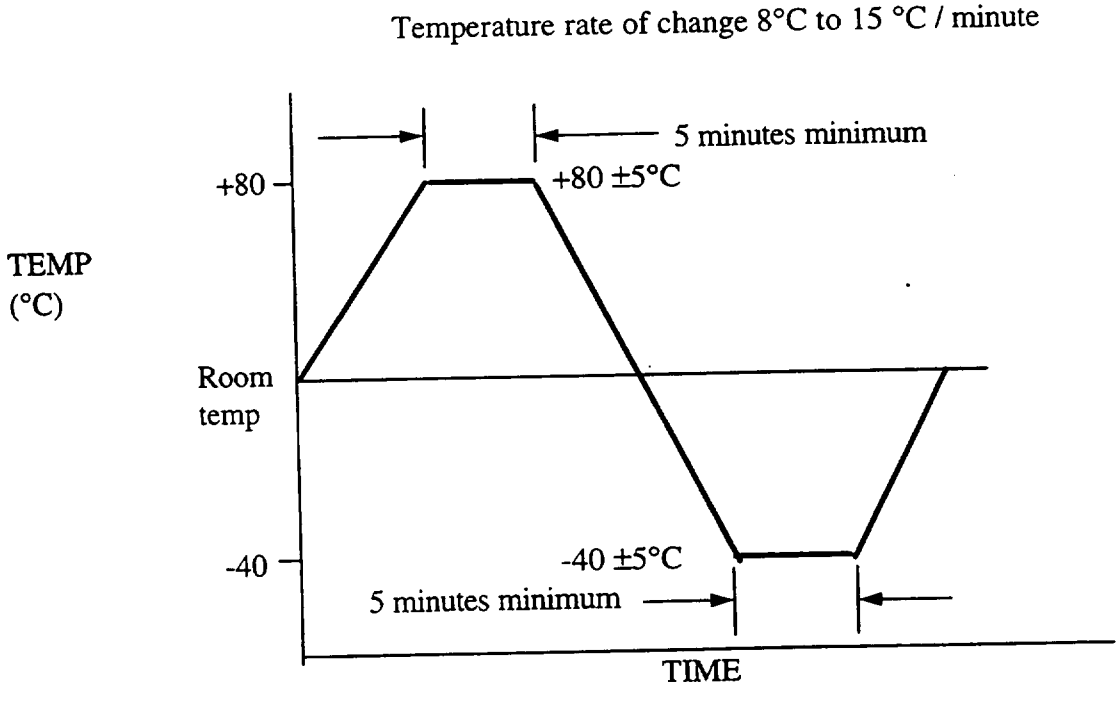
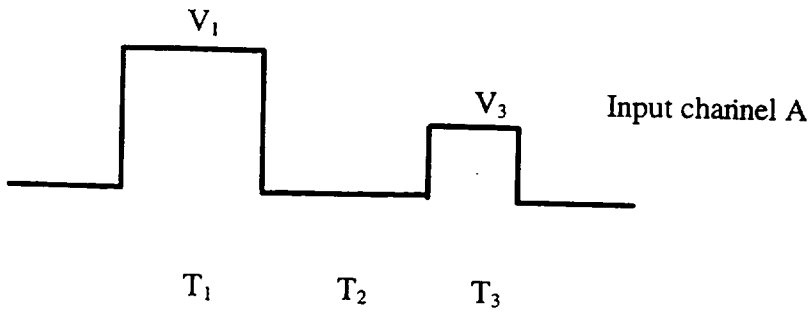


FIGURE 2. One preconditioning cycle.

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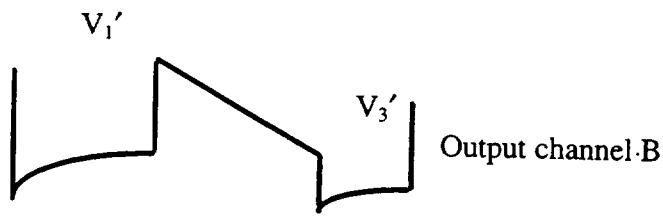
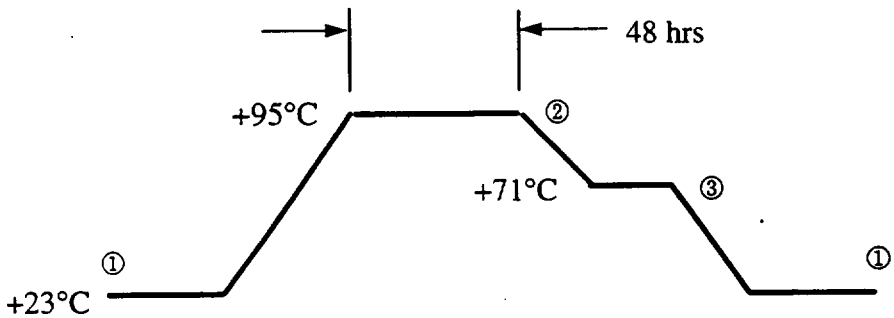


FIGURE 3. Recovery time.

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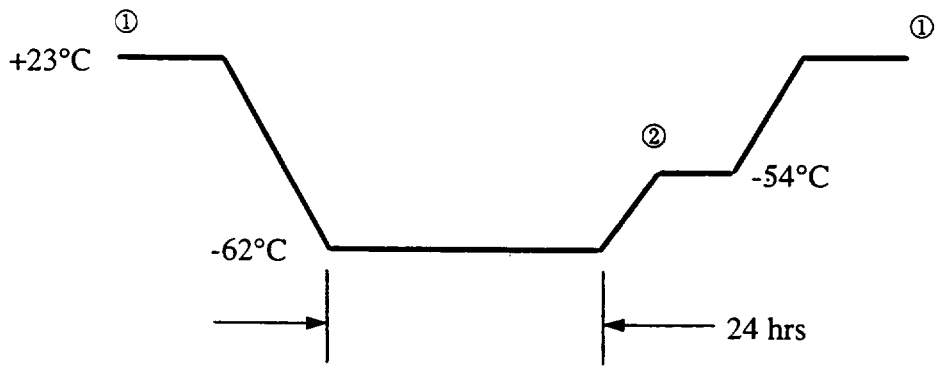
① Voltage gain test (4.5.2) is conducted at these points.

② Test item is operated for 30 minutes at this point.

③ Channel-to-channel tracking (4.5.4) and voltage gain drift tests (4.5.13) are conducted at this point.

FIGURE 4. High temperature profile.

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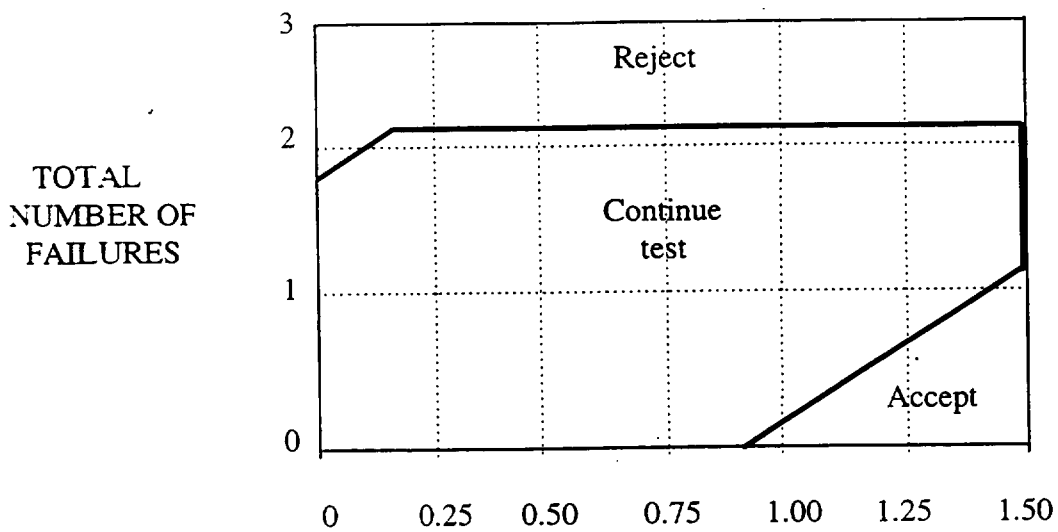


① Voltage gain test (4.5.2) is conducted at these points.

② Channel to channel tracking (4.5.4) and voltage gain drift (4.5.13) tests are conducted at this point

FIGURE 5. Low temperature profile.

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TOTAL TEST TIME*

Number of failures	Reject (equal or less)	Accept (equal or more)
0	NA	0.89
1	NA	1.44
2	0.12	1.50
3	1.50	NA

*Total test time is the total unit hours of equipment "on time" and is expressed in multiples of lower test MTBF, θ_1

FIGURE 6. First article reliability accept - reject criteria.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-49163 B	**2. DOCUMENT DATE (YYMMDD)** 97/27/97

3. DOCUMENT TITLE VIDEO PREAMPLIFIER, AM-6923/UA

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME ROGER LOUISAIRE US Army Communications-Electronics Command	b. TELEPHONE (Include Area Code) (1) Commercial (732) 532-2735 (2) AUTOVON 992-2735
c. ADDRESS (Include Zip Code) ATTN: AMSEL-LC-LEO-E-EP Fort Monmouth, NJ 07703-5023	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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